DIFFERENT SHELLS

CMD Shell:

- It is the default command-line for windows os.
- It provides a text-based interface to interact with the computer, allowing users to execute programs and manage the system through typed commands.
- Used for direct operating system interaction and task automation.
- It does not require any heavy .NET framework to run.
- Its core strength is executing traditional administrative binaries and surviving system failures.

- **Use Cases:**

  - **System Repair & Recovery:** Fixing corrupted hard drives or repairing boot records when Windows won't start.
  - **Rapid Network Diagnostics:** Quickly flushing DNS cache or checking network latency without loading heavy modules.
  - **Legacy Batch Operations:** Running ancient .bat scripts that critical enterprise software still relies on.
  - **Robust File Copying:** Migrating massive amounts of data across drives using built-in, highly resilient tools

POWERSHELL:

- It is a framework that allows cross-platform task automation and configuration management.
- Unlike traditional shells that output text, PowerShell returns .NET objects, making data manipulation and scripting more efficient.
- When you run a command, you aren't getting text; you are getting a programmatic object.
- It features a command-line shell, a scripting language, and a configuration management framework (Desired State Configuration).

- **Use Cases:**

  - **Mass Automation (Active Directory):** Creating 500 new employee user accounts, assigning them to groups, and generating temporary passwords from a single CSV file in seconds.

- **Hardware Auditing:** Querying 50 networked computers to report back their serial numbers, RAM capacity, and BIOS versions.
- **Web API Integration:** Fetching JSON data from a REST API (like a weather service or ticketing system) and converting it directly into a formatted table.
- **Advanced Process Management:** Finding processes consuming the most memory and forcefully terminating them based on specific parameters.

BASH (Bourne again shell):

- Bash's power is derived from the Unix philosophy: using tiny, hyper-focused, incredibly fast tools chained together by the pipeline (|).
- Its true strength is string manipulation.
- It can process gigabytes of raw text or log files in milliseconds, making it the ultimate tool for developers and server administrators.
- **Use Cases:**
    - **Log Analysis:** Sifting through massive Apache or Nginx web server logs to find IP addresses that are attempting to hack the server or causing 404 errors.
    - **CI/CD & DevOps:** Writing the automation scripts that build, test, and deploy software in environments like GitHub Actions, Jenkins, or Docker.
    - **Remote Server Management:** Connecting to headless (no GUI) cloud servers via SSH to update packages, restart services, or manage databases.
    - **Bulk File Modification:** Finding thousands of files matching a specific pattern and renaming them or replacing specific lines of code inside them simultaneously.

ANACONDA PROMPT:

- The Anaconda Prompt's unique power is its **Dependency Solver.**
- In languages like Python and C++, different libraries rely on specific versions of other. Anaconda mathematically solves these complex

dependency trees and builds isolated "sandboxes" (environments) so your projects never break each other.

- Use cases:
  - **Machine Learning Engineering:** Seamlessly switching between a project that uses TensorFlow and another that requires PyTorch, without driver or library collisions.
  - **Scientific Reproducibility:** Exporting an exact blueprint of your environment so another researcher on a different continent can run your code perfectly on their machine.
  - **Data Analysis Workflows:** Bootstrapping a clean environment with Jupyter Notebooks, Pandas, and Matplotlib instantly for a new dataset.

DIFFERENCES BETWEEN BASIC COMMANDS:

Action	**CMD**	**PowerShell**	**Bash**	**Anaconda (Conda)**
List Files	dir	ls or dir	ls	conda list (shows packages)
Change Directory	cd	cd	cd	conda activate [env_name]
Create Directory	mkdir	mkdir	mkdir	conda create -n [name]
Delete File	del	rm or del	rm	conda remove [package]
Copy File	copy	cp or copy	cp	—

Action	CMD	PowerShell	Bash	Anaconda (Conda)
Move/Rename	move	mv or move	mv	—
Clear Screen	cls	cls or clear	clear	cls or clear
Help/Manual	help [cmd]	Get-Help [cmd]	man [cmd]	conda --help